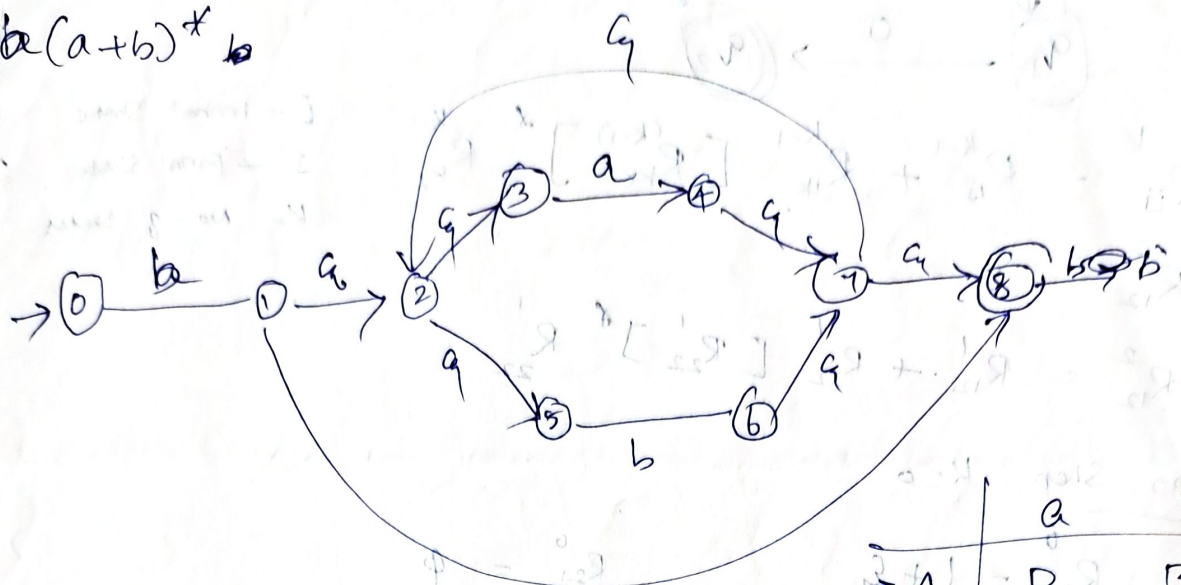


1/ DFA to Regular Expression to DFA

$a(a+b)^*b$



$A \rightarrow$ Starting State

$$A \rightarrow \epsilon\text{-closure}(0) = \{0\}$$

$$\delta(A, a) = \epsilon\text{-closure}(\delta(0, a))$$

$$= \epsilon\text{-closure}(\emptyset) = \emptyset \rightarrow \text{D}$$

	a	b
$\rightarrow A$	D	B
*B	C	E
*C	C	E
*D	D	D
*E	C	E

$$\delta(A, b) = \epsilon\text{-closure}(\delta(0, b))$$

$$= \epsilon\text{-closure}(1)$$

$$= \{1, 2, 3, 5, 8\} \rightarrow \text{B}$$

$$\delta(B, a) = \epsilon\text{-closure}(\delta(\{1, 2, 3, 5, 8\}, a))$$

$$= \epsilon\text{-closure}(\emptyset \cup \emptyset \cup 4 \cup \emptyset \cup \emptyset)$$

$$= \epsilon\text{-closure}(4)$$

$$= \{4, 7, 8, 2, 3, 5\} \rightarrow \text{C}$$

$$\delta(B, b) = \epsilon\text{-closure}(\delta(\{1, 2, 3, 5, 8\}, b))$$

$$= \epsilon\text{-closure}(6)$$

$$= \{6, 7, 8, 2, 3, 5\} \rightarrow \text{E}$$

$$\delta(C, a) = \epsilon\text{-closure}(4) = \text{C}$$

$$\delta(C, b) = \epsilon\text{-closure}(6) = \text{E}$$

$$\delta(E, a) = \epsilon\text{-closure}(4) = \text{C}$$

$$\delta(E, b) = \epsilon\text{-closure}(6) = \text{E}$$

